

A Phenomenological Model of Cosmological Redshift, Type Ia Supernova Time Dilation, and Distance Duality in KUT

— Response Rate of the Dynamic Φ Field Generated by a Phase Transition and Isotropic Phase Transport of Freely Propagating Light —

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4 September 2026

Abstract

This paper considers a cosmological picture in which the universe was formed through a phase transition from a ground-state phase into a non-expanding universe containing a dynamic Φ field, and in which the Φ -response rate $R\Phi(t)$ varies with the elapsed time after the transition. The post-transition dynamic Φ field is the cosmic substrate, with an extremely large effective bulk modulus and mass-equivalent density, and light propagates as a wave of this field. Immediately after the phase transition, the temporal response rate $R\Phi(t)$ of local physical processes was lower than it is today. $R\Phi(t)$ increases over cosmic history, so atomic oscillations, material motion, and radiative processes proceed more rapidly toward the present. With the spatial scale held fixed, this paper describes cosmological redshift and Type Ia supernova time dilation through the common factor $1+z$ produced by the epoch dependence of this response rate. We define the response-rate ratio between emission and observation as $1+z=R\Phi_0/R\Phi_e$ and adopt two transport principles: the locally measured speed of light is the same constant c_0 at every epoch, and the number and temporal ordering of phase fronts in the same finite wave train are conserved during propagation. Under these conditions, the observed time width is multiplied by $1+z$, the frequency by $1/(1+z)$, and the wavelength by $1+z$; the same time dilation therefore appears in Type Ia supernova light curves. Based on the isotropy of the dynamic Φ field, we further propose that the transverse phase scale of freely propagating light is mapped by $1+z$ and that the cross-sectional area of a light bundle is mapped by $(1+z)^2$. Combining photon-energy loss, the reduced arrival rate, and transverse-area growth gives bolometric surface brightness proportional to $(1+z)^{-4}$ and the distance-duality relation $d_L=(1+z)^2 d_A$. Thus redshift, time dilation, surface brightness, and distance duality are linked by a single isotropic light-transport law without invoking spatial expansion. As a minimal example, we examine $R\Phi(t)=t/t_0$. For $t_0=13.8$ Gyr, emission at $z=14$ occurs 0.92 Gyr after the origin of the universe. When past light propagation is compared using the present clock, its effective speed is $c_{\text{eff}}=c_0 R\Phi(t)/R\Phi_0$, while the propagation-time coordinate is transformed at the same time. Treating both consistently cancels the response-rate factor in the distance integral and yields $r=c_0 t_0 z/(1+z)$. With isotropic transport, the linear-response model gives $d_L=c_0 t_0 z(1+z)$, and at low redshift $H_{\text{eff}}=1/t_0$. Unlike a simple tired-light model, this model derives spectral redshift and the observed $1+z$ time dilation from the same physical mechanism. Direct fits to supernova redshifts and brightnesses, and to BAO angles and redshift intervals, can determine the response function and distance curve observationally.

Fitting the generalized distance law $d_L=c_0 t_0 z(1+z)^\gamma$ to 1,575 uncalibrated Pantheon+ distance moduli ($z_{\text{HD}}>0.01$, $z\leq 1.6$) using the full statistical-plus-systematic covariance gives $\gamma=0.63086$, $\Delta M=-0.05216$ mag, and $\chi^2=1391.70$. Under the same conditions and over the same range, flat- Λ CDM gives $\Omega_m=0.33043$ and $\chi^2=1384.80$, a difference of $\Delta\chi^2=6.90$. The reduction from $\Delta\chi^2=15.01$ over the full range $z\leq 2.26137$ shows that the few points at the highest redshifts strongly affect the comparison. For $z\leq 1.6$, the two distance-modulus curves are visually very close on a conventional plot, and the generalized KUT distance law reproduces the distance–redshift relation with an accuracy close to that of standard cosmology. However, γ is a

phenomenological exponent determined from observations and must be distinguished from the parameter-free reference value $\gamma=1$.

Keywords: phase-transition cosmogenesis, redshift, Type Ia supernovae, time dilation, isotropic phase transport, distance-duality relation, Tolman surface brightness, non-expanding universe, cosmic expansion, inflation, Big Bang, BAO

Central claim Most redshift hypotheses that do not invoke cosmic expansion, especially simple tired-light models that assume only a loss of photon energy, cannot account for the observed $1+z$ time dilation of Type Ia supernovae and have therefore been strongly excluded by observations. In contrast, the present model uses the Φ -response-rate ratio, conservation of phase-front number, and isotropic phase transport to describe redshift, time dilation, the Tolman surface-brightness law, and distance duality within a single light-transport mechanism.

1. Introduction

This paper adopts a cosmological picture in which the universe was formed by a phase transition and the spatial scale does not subsequently expand. The analysis does not depend on the microscopic properties of the pre-transition phase, the phase-transition front, energy redistribution, or the details of matter generation. It requires only that the dynamic Φ field became established as the cosmic substrate after the transition and that its response rate changed with the elapsed time since the transition. Cosmic history is therefore described as the evolution of the response rate of the dynamic Φ field rather than as a history of spatial expansion.

The kinematic model of phase-transition-front cosmology that provides the cosmogenic background for this study has been published on SSRN as an earlier paper [11]. Against that background, the present paper formulates a phenomenological model that describes cosmological redshift, the time dilation observed in Type Ia supernova light curves, and the distance-duality relation within a common framework.

The dynamic Φ field is a continuous substrate present throughout the universe. Representative effective properties adopted in KUT are a bulk modulus $K\Phi \approx 10^{37} - 10^{38}$ Pa and a mass-equivalent density $\rho\Phi \approx 10^{20} - 10^{21}$ kg m⁻³. The wave speed $c\Phi \approx \sqrt{(K\Phi/\rho\Phi)}$ determined by their ratio is of the order of the speed of light. In the dynamic phase, the effective viscosity is taken to be nearly zero and the elasticity extremely large, and light is treated as a wave propagating through this substrate. These are effective material properties in KUT and provide the physical starting point for describing gravity, light, and matter in terms of the same substrate.

In the post-transition dynamic Φ field, a common Φ -response rate governing atomic oscillations, material motion, chemical reactions, radiative processes, and other local physical processes increased with the elapsed time since the phase transition. Past clocks are therefore slow when compared with the present standard, and all local physical processes accelerate by the same ratio toward the present. We denote this epoch-dependent response rate by $R\Phi(t)$ and adopt $R\Phi(t)/R\Phi,0 = t/t_0$ as its minimal form.

In standard FLRW cosmology, cosmological redshift is described by a change in the cosmic scale factor. Type Ia supernovae are not only standardizable light sources but also standard clocks whose light-curve widths can be compared. Goldhaber et al. showed that the light curves of distant Type Ia supernovae are stretched by $1+z$ [1]. An analysis of 1,504 Type Ia supernovae from the Dark Energy Survey reported $b = 1.003 \pm 0.005$ (statistical) ± 0.010 (systematic) for $\Delta t_{\text{obs}} = \Delta t_{\text{em}}(1+z)^b$ [2]. Any alternative model of redshift must therefore reproduce not only the frequency decrease of spectral lines but also this time dilation by the same factor.

Many historically proposed non-expanding redshift hypotheses reduce only the energy or frequency of light during propagation and therefore do not stretch an entire supernova light curve by $1+z$. This observation has provided a decisive test that strongly excludes simple tired-light models [1,2]. KUT does not append an auxiliary mechanism merely to evade this test. Instead, it assumes that the same Φ -response-rate ratio that produces redshift maps the temporal structure of both a finite wave train and a light curve by the factor $1+z$.

The longitudinal wavelength, transverse light bundle, amplitude, and arrival time of light should be connected by a single transport law. Because the dynamic Φ field has no preferred direction in KUT, we propose isotropic transport in which the $1+z$ phase stretching observed longitudinally also acts by the same ratio on the transverse phase scale. This single principle yields the distance-duality relation and the Tolman surface-brightness law together with time dilation.

The purpose of this paper is to formulate a phenomenological model in which epoch-dependent changes in the response rate to the dynamic Φ field—the substrate of KUT (Kawase Unified Theory)—describe redshift, time dilation, and light-flux transport within one framework. Without assuming expansion of the spatial scale, and while holding the local speed of light constant at every epoch, the model treats the phase history of freely propagating light continuously from the past to the present in both longitudinal and transverse directions.

We first formulate the Φ -response rate, constancy of the local speed of light, and conservation of phase-front number in freely propagating light as basic principles. We then derive redshift, wavelength stretching, Type Ia supernova time dilation, the time–redshift relation, and light-travel distance. Next, isotropic phase transport is used to derive flux, luminosity distance, angular-diameter distance, distance duality, and the Tolman law, after which methods of testing the model with direct supernova and BAO observables are presented.

2. Model Definition and Physical Assumptions

2.1 Φ -Response Rate

Let t be the physical cosmic time defined by connecting the local clocks at successive epochs from the origin of the universe, and let t_0 denote the present. We write the response rate of local physical processes—such as atomic oscillations, material motion, and radiative processes—to the dynamic Φ field as $R\Phi(t)$. The present value is $R\Phi_0 = R\Phi(t_0)$, and the value at emission time t_e is $R\Phi_e = R\Phi(t_e)$. We assume that $R\Phi(t)$ is smaller in the past and increases toward the present.

The temporal response rate $R\Phi(t)$ used here is the relative rate at which clocks and physical processes proceed at each epoch when compared with present local physical processes. Put simply, it represents the rate of physical change—the rate at which clocks advance. It is not a rate relative to an external absolute time, but a relative rate obtained by comparing different epochs with the present standard. An observer at a given epoch does not perceive time as slow because the observer's clock, body, and measuring instruments all obey the same response rate.

2.2 Constancy of the Local Speed of Light

The speed of light measured by a local observer at any epoch using the atomic clock and ruler of that epoch is always taken to be the same constant c_0 .

$$c_{\text{local}}(t) = c_0 \quad (1)$$

This assumption requires a distinction between the locally measured speed of light and the effective speed obtained when past propagation is compared using the present clock. If T is an auxiliary coordinate that permits comparison using a common present-day clock, while t is physical cosmic time, the former speed is $c_0 R\Phi(t)/R\Phi_0$ whereas the latter is c_0 .

2.3 Phase Transport of Freely Propagating Light

For a finite wave train emitted in the past, we assume that no absorption, re-emission, splitting, creation of new phase fronts, or disappearance of existing phase fronts occurs during propagation. The number N of phase fronts in the same wave train and their ordering are conserved.

$$N_{\text{obs}} = N_e \quad (2)$$

Phase-transport principle Equation (2) is the basic KUT transport principle that freely propagating light neither creates nor destroys phase fronts unless absorption or re-emission occurs. Light generated in the past propagates while preserving the temporal structure and phase ordering at emission and is not reconstructed en route to match the material clocks of successive epochs. If it were reconstructed, new phase fronts would have to be added to stretch the observed time width, violating $N_{\text{obs}}=N_{\text{e}}$. A microscopic formulation using a Maxwell-type equation containing the Φ field, a wave equation, or an eikonal equation is the next step in developing this principle.

The dynamic Φ field is macroscopically isotropic and has no preferred transverse direction for freely propagating light. We extend this isotropy to isotropic phase transport: the same response-rate ratio that maps the longitudinal phase interval also acts on the transverse phase scale. This is distinct from the statement that the electric field is transverse; it is a propagation law governing the cross-sectional area and amplitude of a light bundle.

2.4 Separate Local Waves and One Continuously Propagating Wave Train

Light emitted by an atom in the past according to its local clock and light emitted by an atom today according to the present local clock are separate local waves that are not mutually connected. If local physical constants and atomic standards preserve the same dimensionless relations, the reference wavelengths measured by the respective local observers are equal.

$$\lambda_{\text{local},e} = \lambda_{\text{local},0} = \lambda_0 \quad (3)$$

By contrast, a single wave train created in the past and propagating continuously to the present connects epochs with different response rates. The redshift in this paper is defined as a transport mapping for this second case.

3. Redshift, Frequency, and Wavelength

3.1 Definition of Redshift by the Response-Rate Ratio

We relate the observed redshift to the response-rate ratio between emission and observation as follows.

$$1 + z = \frac{R_{\Phi,0}}{R_{\Phi,e}} \quad (4)$$

Equation (4) is the central relation of this model. Redshift is defined not as a ratio of spatial scales but as the ratio of the Φ -response rates at observation and emission. Once the time evolution of $R\Phi(t)$ is specified, Equation (4) uniquely links observed redshift to emission time and predicts frequency, time width, and distance.

3.2 Conservation of Phase-Front Number in a Finite Wave Train

Let f_e be the local frequency at emission and Δt_e the emission duration. The number of phase fronts in the finite wave train is $N_e = f_e \Delta t_e$. We assume that the response-rate ratio maps the observed duration of the entire wave train.

$$\Delta t_{\text{obs}} = (1 + z) \Delta t_e \quad (5)$$

Using Equations (2) and (5), the observed frequency is obtained by dividing the number of phase fronts by the observed time width.

$$f_{\text{obs}} = \frac{N_{\text{obs}}}{\Delta t_{\text{obs}}} = \frac{f_e}{1+z} \quad (6)$$

Because the local speed of light at the observation point is c_0 , the observed wavelength is

$$\lambda_{\text{obs}} = \frac{c_0}{f_{\text{obs}}} = (1 + z)\lambda_e \quad (7)$$

Thus the time width and period are multiplied by $1+z$, the frequency by $1/(1+z)$, and the wavelength by $1+z$. The same factor therefore links redshift and Type Ia supernova time dilation.

3.3 Consequence of Assuming an Invariant Wavelength

If $\lambda_{\text{obs}} = \lambda_e$ is assumed at the observation point, then $f_{\text{obs}} = f_e$ because the local speed of light is the same c_0 . However, because the observed time width is multiplied by $1+z$, the observed number of phase fronts would be

$$\lambda_{\text{obs}} = \lambda_e \Rightarrow N_{\text{obs}} = (1 + z)N_e \quad (8)$$

At $z=14$, this gives $N_{\text{obs}}=15N_e$ and would require phase creation or splitting during propagation. In the present model, which adopts Equation (2), an invariant wavelength and time dilation cannot both be maintained.

3.4 Predictions from the Basic Principles

Equations (6) and (7) follow directly by combining the time mapping (5), phase-front conservation (2), and constancy of the local speed of light (1). For a single continuous finite wave train, multiplying the observed time width by $1+z$, multiplying the frequency by $1/(1+z)$, and multiplying the wavelength by $1+z$ are therefore not independent assumptions but three representations of the same phase transport. This closed relation distinguishes KUT from tired-light models in which only the frequency changes.

In quantum terms, $E=hf$, so Equation (6) also implies that the observed photon energy is reduced by $1/(1+z)$. The model does not assume cumulative energy loss through scattering; instead, it describes redshift as the phase history of freely propagating light connecting epochs with different Φ -response rates. Expressing the local energy exchange between light and the Φ field in a wave equation remains a task for extending this phenomenology into a microscopic theory.

4. Time Dilation of Type Ia Supernovae

A Type Ia supernova light curve is regarded as a finite temporal structure composed of many emission phases or flux elements arranged in time. If their temporal order is conserved in freely propagating light and the same response-rate ratio acts on every time interval, the entire light curve is mapped by $1+z$ along the time axis while retaining its shape.

$$L_{\text{obs}}(t) \propto L_e \left(\frac{t}{1+z} \right) \quad (9)$$

This relation agrees with the time-dilation factor measured by Goldhaber et al. [1] and by the DES analysis [2]. Whereas a simple tired-light model produces only a frequency decrease, in KUT the same response-rate ratio acts on both the spectrum and the light curve. The model therefore succeeds in describing redshift and Type Ia supernova time dilation within the same phenomenological framework, without assigning them to separate mechanisms. Independent observables—the distance–redshift relation, surface brightness, angular-diameter distance, and BAO—are required to distinguish the model from the standard expanding universe.

Type Ia supernovae also have an intrinsic stretch–luminosity relation between light-curve width and peak luminosity. In fitting KUT to Pantheon+, the cosmological $1+z$ time dilation must be separated from standardization of individual supernova variations. Pantheon+ contains 1,701 light curves for 1,550 Type Ia supernovae [3,4].

5. Linear-Response Model

5.1 Minimal Form

As a minimal example that introduces no extra curvature, inflection point, or relaxation time, we normalize the present response rate to unity and assume that $R\Phi(t)$ increases linearly from the origin of the universe.

$$\frac{R\Phi(t)}{R\Phi,0} = \frac{t}{t_0} \quad (10)$$

Equations (4) and (10) give the following relations among emission time, lookback time, and redshift.

$$1 + z = \frac{t_0}{t_e} \quad (11)$$

$$t_e = \frac{t_0}{1+z} \quad (12)$$

$$t_{\text{lookback}} = \frac{t_0 z}{1+z} \quad (13)$$

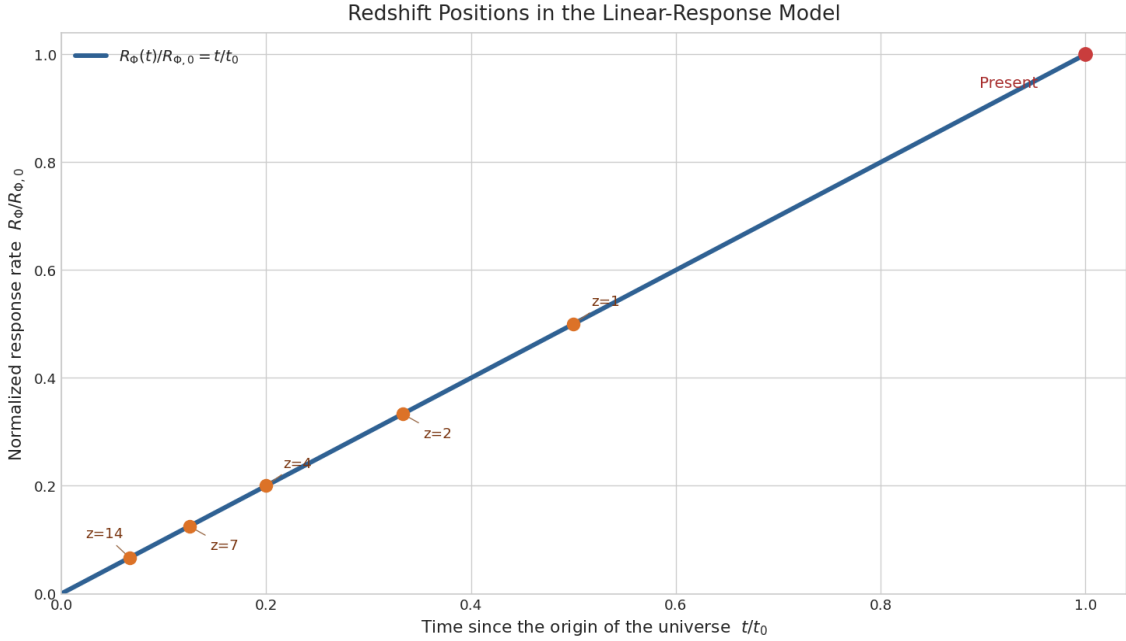


Figure 1. Linear-response model $R\Phi(t)/R\Phi,0=t/t_0$ and the positions of representative redshifts. Points at higher z cluster toward the origin of the universe.

5.2 Numerical Example for $t_0=13.8$ Gyr

Table 1 gives the values for $t_0=13.8$ Gyr. The times scale in proportion to t_0 , whereas the normalized response rate is determined solely by z .

z	$R\Phi,e/R\Phi,0$	Cosmic age at emission (Gyr)	Lookback time (Gyr)
1	0.5000	6.900	6.900
2	0.3333	4.600	9.200
3	0.2500	3.450	10.350
4	0.2000	2.760	11.040
5	0.1667	2.300	11.500
6	0.1429	1.971	11.829
7	0.1250	1.725	12.075
8	0.1111	1.533	12.267
9	0.1000	1.380	12.420

z	$R\Phi_e/R\Phi_0$	Cosmic age at emission (Gyr)	Lookback time (Gyr)
10	0.0909	1.255	12.545
11	0.0833	1.150	12.650
12	0.0769	1.062	12.738
13	0.0714	0.986	12.814
14	0.0667	0.920	12.880

Table 1. Response rates and time correspondence for $z=1-14$ in the linear-response model ($t_0=13.8$ Gyr).

6. Time Coordinates and Light-Travel Distance

The speed of light measured by a local observer at every epoch is c_0 . Past local clocks, however, are slow relative to the present standard, so the effective speed is smaller when the advance of light through fixed space is expressed using the present clock T . Both the speed and the time interval used in the integral must nevertheless be transformed together.

6.1 Physical Cosmic Time and Present-Referenced Time

Let t be physical cosmic time and T the comparison coordinate defined by the present clock. Because $dt/dT=R\Phi(t)/R\Phi_0$ and the local speed of light is $dr/dt=c_0$, the effective speed referenced to the present is

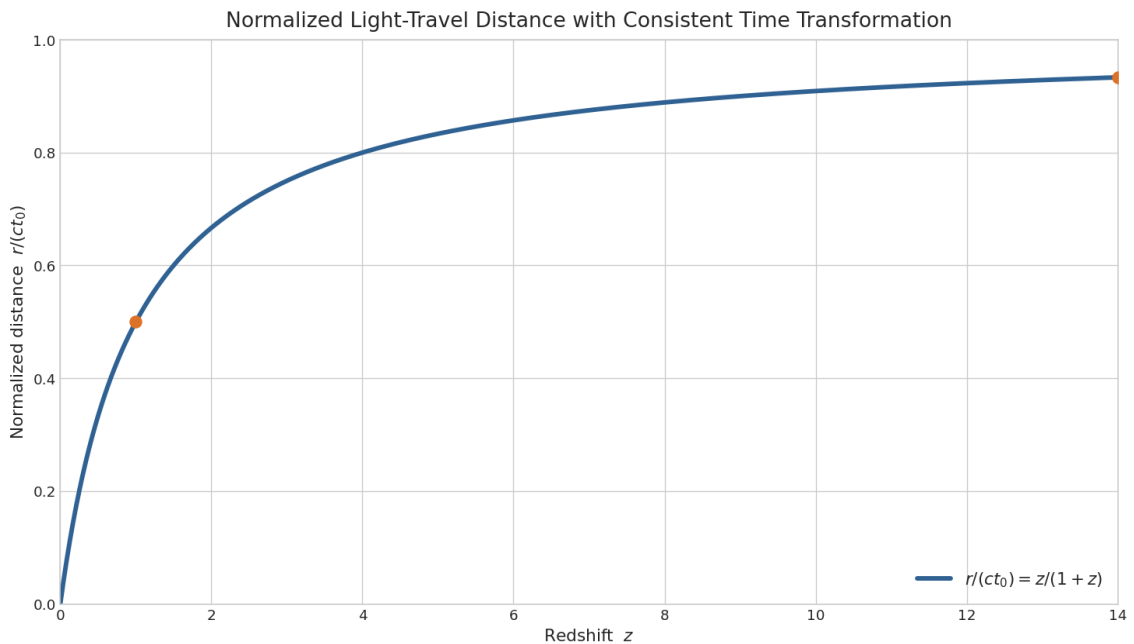
$$\frac{dt}{dT} = \frac{R\Phi(t)}{R\Phi_0}, \quad \frac{dr}{dT} = c_0 \frac{R\Phi(t)}{R\Phi_0} \quad (14)$$

6.2 Distance Integral Invariant under the Time Transformation

Equation (14) gives $R\Phi(t)dT/R\Phi_0=dt$. Consequently, integrating the slower present-referenced effective speed over present-referenced time gives the same distance as integrating the local speed of light over physical cosmic time.

$$r = \int_{T_e}^{T_0} c_0 \frac{R\Phi}{R\Phi_0} dT = c_0(t_0 - t_e) = \frac{c_0 t_0 z}{1+z} \quad (15)$$

For $z=14$ and $t_0=13.8$ Gyr, $r=12.88$ Gly. At high redshift, the distance asymptotically approaches $c_0 t_0$. If the response rate is applied only to the present-referenced effective speed while the integration variable is left as t , the slowing of the same clock is counted twice.

Figure 2. Normalized light-travel distance obtained with a consistent time transformation. It approaches unity at high z .

6.3 Low-Redshift Slope

The low-redshift limit of Equation (15) is $r \approx c_0 t_0 z$. Comparison with the standard Hubble law $r \approx c_0 z / H_0$ gives an apparent low-redshift coefficient $H_{\text{eff}} = 1/t_0$ in this model. Fixing $t_0 = 13.8$ Gyr uniquely gives $H_{\text{eff}} \approx 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$, comparable to the observed low-redshift Hubble-law slope. At higher redshift, the curvature is a model-specific prediction that can be tested with supernova and BAO observations.

Avoiding double counting The variable t is already physical cosmic time defined by connecting local clocks. The response rate must therefore not be applied a second time to $\int c_0 dt$. When the present reference is used, both the speed and the time interval must be transformed to T .

7. Isotropic Phase Transport, Flux, and Distance Duality

7.1 Transverse Phase Scale

Equation (7) shows that the longitudinal phase interval of the same wave train is multiplied by $1+z$. If the dynamic Φ field is macroscopically isotropic, the same transport mapping is not restricted to the direction of propagation but also acts on the transverse phase scale of the light bundle. We adopt as the isotropic phase-transport principle that a representative transverse linear scale b obeys

$$b_{\text{obs}} = (1 + z)b_e \quad (16)$$

The cross-sectional area A of a light bundle has two transverse degrees of freedom and therefore changes as follows.

$$A_{\text{obs}} = (1 + z)^2 A_e \quad (17)$$

Transverse stretching does not follow mathematically from wavelength stretching alone. KUT nevertheless treats them not as separate corrections but as the longitudinal and transverse components of one phase transport through an isotropic dynamic Φ field.

7.2 Flux and Luminosity Distance

Consider the bolometric flux received at distance r from a source with intrinsic luminosity L . Geometrical spherical spreading in fixed space gives $1/(4\pi r^2)$. Combining this with the factors $1/(1+z)$ for observed photon energy, $1/(1+z)$ for arrival rate, and $1/(1+z)^2$ for the transverse area from Equation (17) gives

$$F = \frac{L}{4\pi r^2 (1+z)^4} \quad (18)$$

Defining the luminosity distance by $F = L/(4\pi d_L^2)$ and taking the angular-diameter distance in fixed space to be $d_A = r$ gives

$$d_A = r, \quad d_L = (1 + z)^2 r \quad (19)$$

Even with the spatial scale held fixed, including transverse phase transport of freely propagating light connects apparent brightness and apparent angle through the same propagation law.

7.3 Distance Duality and Tolman Surface Brightness

Equation (19) immediately gives the following relation between luminosity distance and angular-diameter distance.

$$d_L = (1 + z)^2 d_A \quad (20)$$

This has the same form as Etherington's distance-duality relation [5,6]. A recent model-independent analysis using DESI DR2 angular-diameter distances and several Type Ia supernova samples is also statistically consistent with this relation [8]. In standard cosmology the relation follows from expansion of the spacetime metric and photon-number conservation; KUT obtains the same observational relation from the Φ -response-rate ratio and isotropic phase transport of freely propagating light.

Expressing the same result as the bolometric surface brightness of the emitting surface gives

$$\mathcal{S}_{\text{obs}} = \frac{\mathcal{S}_e}{(1+z)^4} \quad (21)$$

which reproduces the Tolman surface-brightness law [9]. Equations (20) and (21) are not two independent adjustments but two representations of the same transport law combining photon energy, arrival rate, and transverse area. The model thus links time dilation, distance duality, and surface-brightness dimming through a common $1+z$ mapping.

8. Distance–Redshift Curves and Direct Observables

8.1 Quantities Measured by Type Ia Supernovae

Direct observables for Type Ia supernovae are the spectroscopic redshift z , the flux F received at Earth, color, and light-curve shape. Standardizing the supernovae and calibrating the intrinsic luminosity L or absolute magnitude M yields the luminosity distance and distance modulus.

$$F = \frac{L}{4\pi d_L^2}, \quad \mu = 5\log_{10}\left(\frac{d_L}{\text{Mpc}}\right) + 25 \quad (22)$$

Absolute supernova distances therefore depend on the calibration of M , but the shape of the brightness–redshift curve is strongly constrained by many supernovae. A KUT test should not use distances fixed after conversion under Λ CDM. Instead, the distance modulus calculated from Equation (19), Equation (24), or the generalized distance law in Section 8.4 should be fitted directly to the Pantheon+ observables and their full covariance matrix, with M as a common calibration parameter [3,4].

8.2 Quantities Measured by BAO

BAO analyses use the statistical distribution of many galaxies and quasars to measure the apparent transverse angle θ_{BAO} and line-of-sight redshift interval Δz_{BAO} corresponding to a reference length r_d . Representative relations are

$$\theta_{\text{BAO}} = \frac{r_d}{D_M}, \quad \Delta z_{\text{BAO}} = \frac{H_{\parallel}(z)r_d}{c_0} \quad (23)$$

The observations directly constrain D_M/r_d and $H_{\parallel}r_d$, or equivalently D_H/r_d . DESI DR2 measured BAO scales at several redshifts using more than 14 million galaxies and quasars [7]. Because the early-universe physics of KUT may differ from that of the standard model, r_d calibrated under standard cosmology should not be fixed without qualification; it must be derived as the KUT reference scale or fitted simultaneously. BAO is not itself the mechanism that produces redshift; it is an independent standard ruler that tests the redshift–distance relation predicted by KUT without relying on luminosity.

8.3 Distance Curve of the Linear-Response Model

For the linear-response model, Equation (15), $r = c_0 z/(1+z)$, together with Equation (19) gives the luminosity distance

$$d_L(z) = c_0 t_0 z(1+z) \quad (24)$$

At low redshift, $d_L \approx c_0 t_0 z$, and $t_0 = 13.8$ Gyr gives $H_{\text{eff}} = 1/t_0 \approx 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$. At high redshift, the curvature $z(1+z)$ appears. This is the model's reference prediction and can be tested directly with Pantheon+ and BAO. Hereafter, Equation (24) with $\gamma=1$ is called the parameter-free reference law; Section 8.4 tests the curvature required by Pantheon+.

The linear form is the first candidate with the fewest free parameters. If observations require different curvature, the distance law should first be generalized phenomenologically to measure the required curvature. One must then identify theoretically whether that curvature arises from $R\Phi(t)$ or from the light-bundle transport law. Section 8.4 performs this first step using a constant exponent γ .

8.4 Phenomenological Test of the Distance Curve with Pantheon+ ($z \leq 1.6$)

At the high-redshift end, the number of observations is small, and the effects of calibration, selection, and standardization corrections may become relatively more important. We therefore restrict the primary Pantheon+ comparison in this paper to $z \leq 1.6$. The parameter-free form of Equation (24) is the $\gamma=1$ reference prediction corresponding to isotropic phase transport. To measure the curvature required by Pantheon+, we phenomenologically generalize the luminosity distance to $d_L^{\text{KUT}}(z) = c_0 t_0 z(1+z)^\gamma$ and fix $t_0 = 13.8$ Gyr. The exponent γ and the brightness zero point ΔM are fitted simultaneously.

From the public Pantheon+ file Pantheon+SH0ES.dat, we selected 1,575 uncalibrated distance-modulus entries satisfying $\text{IS_CALIBRATOR}=0$, $z_{\text{HD}} > 0.01$, and $z \leq 1.6$ and used the corresponding statistical-plus-systematic covariance submatrix from Pantheon+SH0ES_STAT+SYS.cov. Refitting the generalized KUT law gives $\gamma = 0.63086$, $\Delta M = -0.05216$ mag, and $\chi^2_{\text{KUT}} = 1391.70$. Refitting flat- Λ CDM under the same conditions to the same 1,575 entries gives $\Omega_m = 0.33043$ and $\chi^2_{\Lambda\text{CDM}} = 1384.80$.

Thus $\Delta\chi^2 = \chi^2_{\text{KUT}} - \chi^2_{\Lambda\text{CDM}} = 6.90$. By comparison, using all 1,580 entries over $z \leq 2.26137$ gives $\Delta\chi^2 = 15.01$; restricting the analysis to $z \leq 1.6$ reduces the difference to 6.90. The five excluded entries at $z > 1.6$ and their covariance make a large contribution to the full-range model difference. Accordingly, this paper uses $z \leq 1.6$ as the primary analysis and treats the full-range result as a sensitivity comparison. Over the primary range, generalized KUT has a χ^2 larger by 6.90 than flat- Λ CDM, but the distance-modulus curves are visually very close on a conventional plot, and the generalized law reproduces the observed curve with an accuracy close to that of standard cosmology.

The value $\gamma = 0.63086$ is fitted from the primary $z \leq 1.6$ Pantheon+ analysis; it is not a theoretical prediction independently derived from a Φ -field wave equation. The result shows that the generalized KUT law, with a single curvature exponent, can closely approach the supernova distance–redshift curve; it does not mean that the parameter-free KUT reference law $\gamma=1$ is supported.

Furthermore, if $d_A = r$ and Equation (15) are retained while the generalized distance law is used, then $d_L = (1+z)^{\gamma+1} d_A$. At the best-fit $\gamma = 0.63086$, the exponent is 1.63086, inconsistent with the exponent 2 required by the distance-duality relation in Equation (20). Under the same definitions, the corresponding bolometric surface-brightness exponent is $-2(\gamma+1) = -3.26172$, also inconsistent with -4 in Equation (21). Therefore, making the generalized supernova fit, distance duality, and the Tolman law hold simultaneously requires at least one of $R\Phi(t)$, the angular-diameter distance, or the transverse light-bundle transport law to be reformulated on the basis of additional observations. Related nonstandard fits to Pantheon+ include the covarying-coupling-constants (CCC) and CCC+TL hybrid models proposed by Gupta [12]. Their redshift decomposition and distance law differ from the response-rate and isotropic-transport construction used here.

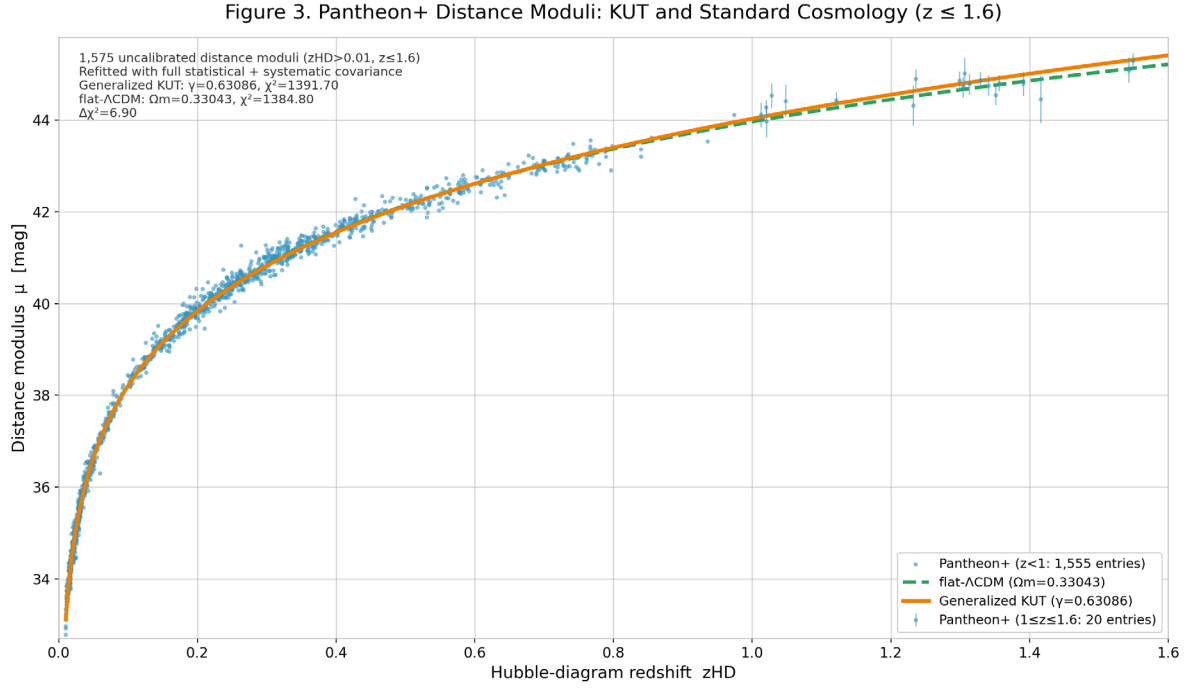


Figure 3. Pantheon+ distance moduli (primary analysis range $z \leq 1.6$; 1,575 uncalibrated distance-modulus entries). The generalized KUT fit gives $\gamma=0.63086$ and $\chi^2=1391.70$, whereas flat- Λ CDM gives $\Omega_m=0.33043$ and $\chi^2=1384.80$, so $\Delta\chi^2=6.90$. The blue circular points show all 1,575 entries; error bars are added to the 20 entries at $1 \leq z \leq 1.6$. The displayed error bars use the diagonal terms only, whereas the fit uses the full statistical-plus-systematic covariance submatrix.

8.5 Why the Distance Curve Arises in Standard Cosmology

In a spatially flat standard FLRW cosmology, the expansion rate $H(z)$ changes with epoch according to the densities of matter, radiation, and dark energy. Schematically,

$$H(z) = H_0 \sqrt{\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_\Lambda} \quad (25)$$

and the comoving distance is the integral of the expansion history,

$$\chi(z) = c_0 \int_0^z \frac{dz'}{H(z')} \quad (26)$$

The angular-diameter and luminosity distances are therefore given by [6]

$$d_A^{\text{FLRW}} = \frac{\chi}{1+z}, \quad d_L^{\text{FLRW}} = (1+z)\chi \quad (27)$$

The standard-model distance curve is not linear because $H(z)$ is not constant and the expansion rate is integrated over every epoch traversed by the light. KUT does not use this expansion history; it constructs the distance curve from the evolution of $R\Phi(t)$ and isotropic light transport. The two models should be compared using direct observables—supernova brightnesses and BAO angles and redshift intervals—not only distances obtained after model-dependent conversion.

9. Physical Cosmic Age at $z=14$

In the linear model, $t_e = t_0/15$ at $z=14$. For $t_0=13.8$ Gyr, the physical cosmic age at emission is 0.92 Gyr, or 920 million years. Because t is defined here as physical cosmic time obtained by connecting local clocks at successive epochs, this 0.92 Gyr is used as the time available for local physical processes to have proceeded by that epoch.

Multiplying by $R\Phi(t)/R\Phi,0$ and integrating once more would count again the clock slowing already included in t . We therefore do not perform an additional response-rate integral and adopt 0.92 Gyr as the physical cosmic age.

JWST/JADES spectroscopically confirmed galaxies at $z=14.32$ and 13.90 and reported that, under the standard-cosmology conversion, luminous galaxies existed about 300 million years after the Big Bang [10]. The present model assigns a physical cosmic age of about 900 million years to a comparable redshift, providing roughly three times as much time for galaxy formation as the standard conversion. This is a consequence of adopting the present model and may alleviate the problem of rapid early-galaxy formation. It can be tested quantitatively through galaxy-formation calculations using stellar mass, metallicity, and star-formation rate.

10. Observational Distinction from Standard Cosmology

In standard FLRW cosmology, $1+z=a_0/a_e$, and the same factor appears in wavelength stretching and time dilation. KUT uses $R\Phi,0/R\Phi,e$ instead of the scale factor. Because the two descriptions can give the same kinematic relations when only redshift and time dilation are considered, the following independent observables are required to distinguish them.

Discriminating observable	Prediction and test in this model
Pantheon+ luminosity distance	Calculate the distance modulus from Equation (22), Equation (24), and the generalized distance law in Section 8.4, and fit it directly to the 1,575 entries at $z \leq 1.6$ using the full covariance, with absolute magnitude M as the common calibration quantity. Report the $\gamma=1$ reference law separately from the observationally fitted value $\gamma=0.63086$, and treat the full range as a sensitivity comparison.
Distance duality and surface brightness	Compare Equations (20) and (21) with DESI DR2–supernova distance-duality tests and with Tolman tests corrected for galaxy evolution.
BAO	Fit D_M/r_d and $H_{\parallel} r_d$ from Equation (23) simultaneously to transverse and line-of-sight observations and determine the KUT reference length r_d .
Redshift drift	Predict from the present gradient of $R\Phi(t)$ the rate at which z of the same object changes with observing time.
High- z galaxies	Use the physical cosmic age in Equation (12) to predict star-formation histories, stellar masses, and metallicities, and compare them with observations.
Wave propagation	Quantify the conditions under which polarization, spectral-line width, dispersion, phase coherence, and the transverse light bundle are preserved over long-distance propagation.

Table 2. Principal discriminating predictions and tests of the KUT redshift model.

11. Discussion

A strength of the model is that it does not divide redshift and Type Ia supernova time dilation between separate mechanisms; both are described by the same response-rate ratio and the transport condition for one wave train. By holding the locally measured speed of light at c_0 at every epoch while producing differences only in observables that connect the past and present, the construction avoids a direct contradiction with local measurements of the speed of light.

The five basic relations of this paper—the response-rate definition of redshift, mapping of temporal structure, conservation of phase-front number, constancy of the local speed of light, and isotropic phase transport—form a closed phenomenological system linking frequency, wavelength, supernova time width, light-bundle cross-sectional area, surface brightness, and the two cosmological distances. In particular, the reductio argument that an unstretched wavelength would increase the phase-front number by $1+z$ clearly shows why wavelength stretching is required by continuity of the same wave train.

With isotropic phase transport, KUT reproduces $d_L = (1+z)^2 d_A$ and the Tolman-type $(1+z)^{-4}$ surface-brightness law through a physical cause different from that in standard cosmology. These two observational relations are not fitted independently; both follow from one transverse-area law. The linear-response model additionally has the specific reference predictions $d_L = c_0 z(1+z)$, $H_{\text{eff}} \approx 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at low z , and a physical cosmic age of 0.92 Gyr at $z=14$.

The central task in developing the model further is to formulate conservation of phase-front number and the transverse-area law as a Φ -field wave equation or an eikonal transport equation. The value $\gamma=0.63086$ obtained from the primary $z \leq 1.6$ Pantheon+ analysis is a phenomenological measure of the distance curvature required by the observations; it is not derived from the Φ field. Moreover, because it is inconsistent with the $\gamma=1$ distance-duality prediction, γ —or the corresponding light-bundle area law—must be predicted theoretically and independently of the supernova data. The same framework must then be tested for its ability to reproduce BAO, angular-diameter distances, and surface brightness.

12. Conclusions

The KUT redshift model assumes that the universe was formed by a phase transition, that its spatial scale did not subsequently expand, and that the Φ -response rate increased with the elapsed time since the transition. Under this minimal physical picture, combining $1+z=R\Phi_0/R\Phi_e$, constancy of the local speed of light, conservation of phase-front number in the same finite wave train, and the time mapping $\Delta t_{\text{obs}} = (1+z)\Delta t_e$ gives $f_{\text{obs}} = f_e/(1+z)$ and $\lambda_{\text{obs}} = (1+z)\lambda_e$. The model thereby shows that spectral redshift and the $1+z$ time dilation of Type Ia supernovae can be reproduced within the same phenomenological framework based on a single physical mechanism.

Extending the isotropy of the dynamic Φ field to the transverse direction of freely propagating light multiplies the transverse linear scale by $1+z$ and the light-bundle cross-sectional area by $(1+z)^2$. Together with the reductions in photon energy and arrival rate, this gives $F = L/[4\pi r^2(1+z)^4]$, $d_L = (1+z)^2 d_A$, and $S_{\text{obs}} = S_e/(1+z)^4$ simultaneously. Through this single isotropic transport law, the model links supernova time dilation, the distance-duality relation, and the Tolman surface-brightness law.

In the linear-response model $R\Phi(t) = R\Phi_0 t/t_0$, $t_e = t_0/(1+z)$. For $t_0 = 13.8$ Gyr and $z=14$, this gives a physical cosmic age of 0.92 Gyr. Applying the time transformation consistently to both speed and time interval gives a light-travel distance of 12.88 Gly under the same conditions and a luminosity distance $d_L = c_0 z(1+z)$. At low z the model gives $H_{\text{eff}} \approx 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$; the high- z curvature can be tested clearly with supernovae and BAO.

A principal result of the model is that it reproduces the Type Ia supernova time dilation not explained by simple tired-light hypotheses using the same factor as redshift, and further derives the observed distance-duality relation and Tolman law from the same transport mechanism. Simultaneous fits to Pantheon+, DESI BAO, and high-redshift galaxies can quantitatively test the KUT response-rate cosmology without spatial expansion.

In the full-covariance comparison over the primary Pantheon+ range $z \leq 1.6$, generalized KUT gives $\gamma=0.63086$ and $\chi^2=1391.70$, whereas flat- Λ CDM gives $\Omega_m=0.33043$ and $\chi^2=1384.80$, a difference of $\Delta\chi^2=6.90$. The difference is smaller than the full-range value $\Delta\chi^2=15.01$, and the two curves are visually very close in a conventional distance-modulus plot. On the other hand, the best-fit value $\gamma=0.63086$ does not match the reference value $\gamma=1$, which gives distance duality and the Tolman law simultaneously. The generalized formula is a useful phenomenological description, but it does not mean that the isotropic phase-transport law of KUT is observationally complete.

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Appendix A. Logical Relationships among the Principal Equations

Category	Relation	Meaning
Fundamental principle	$1 + z = \frac{R_{\Phi,0}}{R_{\Phi,e}}$	Correspondence between redshift and the response-rate ratio
Fundamental principle	$\Delta t_{\text{obs}} = (1 + z)\Delta t_e$	Mapping of temporal structure
Fundamental principle	$N_{\text{obs}} = N_e$	Conservation of phase-front number
Fundamental principle	$c_{\text{local}} = c_0$	Constant local speed of light at every epoch
Fundamental principle	$\frac{dt}{dT} = \frac{R_{\Phi}(t)}{R_{\Phi,0}}$	Relation between physical cosmic time and present-referenced time
Derived result	$f_{\text{obs}} = \frac{f_e}{1 + z}$	Obtained from the time width and phase-front number
Derived result	$\lambda_{\text{obs}} = (1 + z)\lambda_e$	Obtained from c_0 and the frequency at observation
Derived result	$r = \frac{c_0 t_0 z}{1 + z}$	Integrate speed and time interval in the same coordinate
Fundamental principle	$b_{\text{obs}} = (1 + z)b_e$	The transverse phase scale of freely propagating light also stretches isotropically
Derived result	$A_{\text{obs}} = (1 + z)^2 A_e$	The light-bundle cross-sectional area follows from the transverse linear scale
Derived result	$d_L = (1 + z)^2 d_A$	Reproduces the distance-duality relation
Derived result	$\mathcal{S}_{\text{obs}} = \mathcal{S}_e (1 + z)^{-4}$	Reproduces the Tolman surface-brightness law

Table A1. Basic principles, derived results, and observational predictions of the KUT redshift model.

Note: Table A1 summarizes the parameter-free isotropic phase-transport model with $\gamma=1$. If the generalized distance law with $\gamma=0.63086$ obtained from the primary $z \leq 1.6$ Pantheon+ analysis is adopted, $d_L = (1+z)^2 d_A$ and $\mathcal{S}_{\text{obs}} = \mathcal{S}_e (1+z)^{-4}$ do not follow automatically.